THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 7: **Measurement of Time and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The motion of the tip of a vibrating guitar string is: 1

(a) to-and-fro periodic **(b)** circular
(c) uniform linear **(d)** rest

2. A pendulum clock runs fast. A mechanic wishes to correct it. He should: 1

(a) reduce the bob's mass **(b)** stop the clock
(c) shorten the pendulum **(d)** lengthen the pendulum slightly

3. A graph is a straight line sloping down. For distance-time graphs this is: 1

(a) not possible for distance moved **(b)** rest
(c) possible **(d)** uniform speed

4. The motion of the moon around the Earth takes about: 1

(a) 60 days **(b)** 1 day **(c)** 27 days **(d)** 365 days

5. If a body covers distances in the ratio 1 : 3 : 5 in successive equal seconds, the motion is: 1

(a) uniform **(b)** non-uniform **(c)** at rest **(d)** periodic

6. The reading of a stopwatch is 1 min 25 s. In seconds it is: 1

(a) 85 s **(b)** 65 s **(c)** 145 s **(d)** 125 s

7. The Earth's rotation period is about: 1

(a) 12 h **(b)** 24 h **(c)** 365 days **(d)** 27 days

8. Which of these quantities is measured in m/s? 1

(a) time period **(b)** time **(c)** speed **(d)** distance

9. **Assertion (A):** A straight line sloping upward through the origin in a distance-time graph shows uniform speed. 1

Reason (R): Equal distances are covered in equal intervals of time.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** An odometer measures the distance travelled by a vehicle. 1
Reason (R): A speedometer measures the speed of a vehicle.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A pendulum takes 40 s for 25 oscillations. Find its time period. 2

12. A taxi has a meter that charges for distance. Starting reading 4500 km, end 4528 km. Find the distance and the time at 56 km/h. 2

13. Express a speed of 340 m/s in km/h. 2

14. How would you check if a toy car moves uniformly? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A pendulum clock loses 10 minutes in a day. How many oscillations does it lose if its period is 1 s? 3

16. Light takes about 8 minutes to reach Earth. Find the Sun's distance if light travels 3 x 10^5 km/s. 3

17. Why does an electric toothbrush and a pendulum both show periodic motion? 3

OR

A swimmer swims 50 m in 40 s one way and returns in 50 s. Find average speed for the round trip. 3

18. Explain why the speed of a falling stone increases but the motion of a pendulum is periodic. 3

OR

A scooter moves at 36 km/h for 30 minutes. Find the distance covered in km and in m. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Rohan cycles to the market. He covers 4 km in the first 20 minutes, stops at a shop for 10 minutes, and then covers the remaining 6 km in the next 20 minutes. His sister draws a distance-time graph of his journey. 5
(i) Find his speed in the first 20 minutes in km/h.
(ii) What will the graph look like during the 10-minute halt?
(iii) Find his speed in the last 20 minutes in km/h.
(iv) Find his average speed for the whole journey.

20. Describe the history of timekeeping in India, naming ancient methods and their principles. 5

OR

Explain why motion is relative and how speed can be different for different observers. 5

END OF QUESTION PAPER — Total: 40 Marks